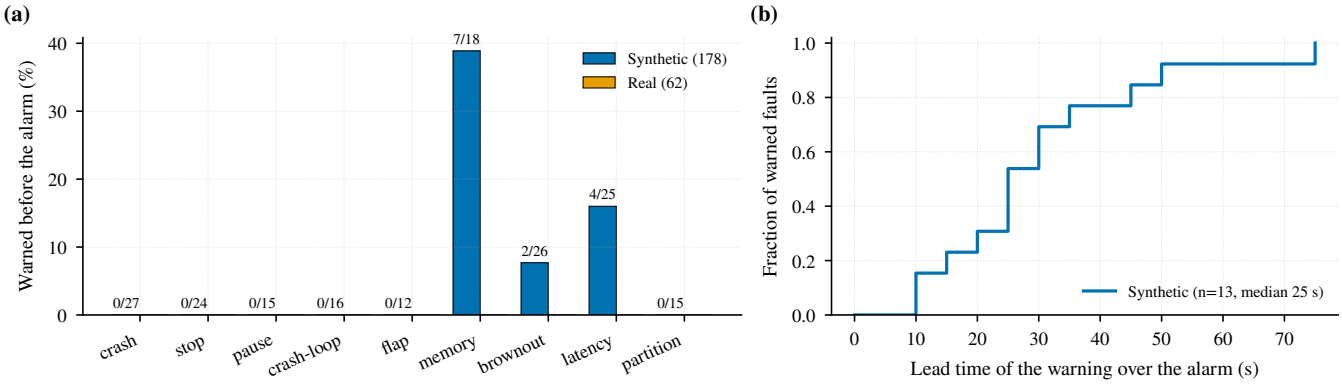
(a)
Warned before the alarm (%)
40
30
20
10
0
Synthetic (178)
Real (62)
0/27
0/24
0/15
0/16
0/12
7/18
2/26
4/25
0/15
crash
stop
pause
crash-loop
flap
memory
brownout
latency
partition
(b)
Fraction of warned faults
1.0
0.8
0.6
0.4
0.2
0.0
Synthetic (n=13, median 25 s)
0
10
20
30
40
50
60
70
Lead time of the warning over the alarm (s)